

DeepScientist-style Hypothesis and Citation Verification Agent

PDF input -> literature retrieval -> research idea cards -> Green / Yellow / Red citation verification

Problem

- LLM research agents can generate plausible ideas with weak or fabricated citations.
- The project checks whether cited literature exists, matches metadata, and supports each claim.
- The grading evidence is the auditable tool chain, not only a chat answer.

Pipeline

- Read first PDF pages and extract title, abstract, keywords, and research problem.
- Generate initial and follow-up search queries.
- Retrieve literature through Crossref and OpenAlex wrappers.
- Generate structured hypothesis cards and verify citations.

Self-built Tools

- PDF reader, query planner, literature searcher, evidence store, hypothesis generator, verifier, report writer, QQ command adapter.
- Every run writes tool_calls.jsonl, retrieved_literature.jsonl, citation_verification.csv, and final_report.md.

Label Rules

- Green: exists, metadata matches, and abstract or snippet directly supports the claim.
- Yellow: exists, but support is partial, weak, abstract-only, or needs human inspection.
- Red: missing paper, invalid DOI, metadata mismatch, or unsupported claim.

Demo Results

- Success case: submission/evidence/success_case, observed Green 3, Yellow 1, Red 0.
- Boundary case: submission/evidence/boundary_case, observed Green 0, Yellow 1, Red 1.
- The Red row is intentionally invalid and demonstrates rejection of bad citations.

Reproducibility

- Run python -m pytest -q.
- Run workflow.py on success_demo.pdf and boundary_demo.pdf.
- Inspect tool_calls.jsonl, retrieved_literature.jsonl, citation_verification.csv, and final_report.md.

Core message

The system favors conservative, auditable verification: uncertain citation support stays Yellow instead of becoming false Green.